Agustín Avelino Pineda

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PROFESSIONAL SUMMARY

Mechatronics Engineering student specializing in embedded systems, power electronics, and C++/Arduino programming. I lead technical teams in electric vehicle competitions.

EDUCATION

Tecnológico de Monterrey, Campus Santa Fe — Santa Fe, CDMX
B.S. in Mechatronics Engineering — Expected Graduation: August 2028

CECyT 9 "Juan de Dios Bátiz" — IPN — Mexico City
Technical Degree in Programming — March 2025

EXPERIENCE

Line-Following AGV with Load/Unload System — Santa Fe, CDMX
Engineering Project — Feb-Jun 2026

- **Designed and implemented** a PD controller with dead zone and exponential smoothing filter on a PWM signal, achieving stable autonomous navigation on a differential-drive robot with BTS7960 drivers.
- Developed the complete logic in C++: line following, trajectory-loss recovery, and station detection, iterating over 20 firmware versions until achieving a robust and functional system.
- Integrated an HX711 load cell with a Kalman filter, quadrature encoders (494 PPR), TMC2209 stepper motor for automated lifting, and an I2C bus at 400 kHz with PCF8574 and OLED on the same bus.

LEADERSHIP & ACTIVITIES

Omega Lightning — Student Racing Team, Tecnológico de Monterrey — Santa Fe, CDMX
Electronics Lead — February 2026 – Present

- Coordinate the electrical development of a competitive kart, leading a 4-person team in task assignment and technical decision-making.
- Redesigned the kart's wiring harness (3/0 to 6 AWG silicone cable), reducing electrical system weight, and built a weatherproof enclosure to protect components against rain and solar radiation.
- Implemented the Safe-to-Touch system to ensure chassis electrical isolation, mitigating shock risks in high-power systems (51.2V / 300A peak).
- Integrated solar panels for continuous autonomous charging of the secondary system and developed an adaptive audio system based on vehicle speed, improving pedestrian safety in the pit area.

Electronics Co-Lead — August 2025 – December 2025

- Supported the design and integration of the kart's electrical system, laying the groundwork for the project I now lead.
- Installed the complete lighting system (front, rear, and brake lights) with a parallel circuit arrangement, ensuring 100% lighting redundancy against failures.
- Collaborated on motor integration, battery systems, and general technical development under competition deadlines.

SKILLS & INTERESTS

Technical: C++, Python, Git/GitHub, Linux, Arduino

Tools: SolidWorks, NX Associate

Languages: Spanish (native), English (B2)

Interests: Robotics, AI, embedded systems, EV racing

Lab Skills: Soldering, electronic systems integration